

CAPACITOR AND INDUCTOR

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Introduction

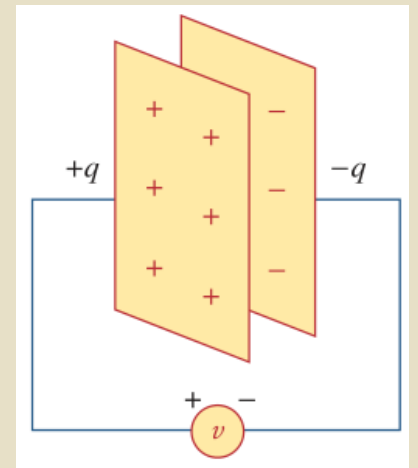
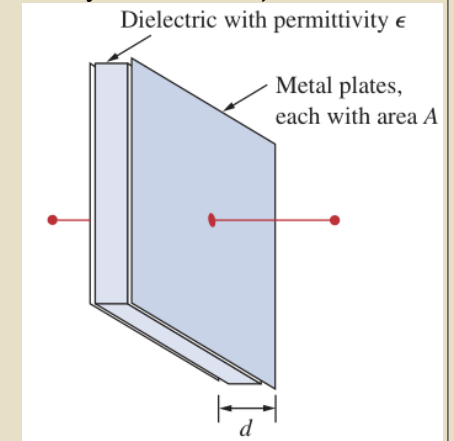
- Passive elements – Inductor, capacitor and Resistor.
- Resistors dissipate energy but inductors and capacitors do not dissipate energy, they store energy. This stored energy can be retrieved later. Thus capacitors and inductors are called storage elements.
- Practical circuits have all of these three characteristics.

Capacitors

- A passive element that stores energy in the form of electric field.
- Uses: Electronics, Communications (i.e. tuning circuit of radio receiver), Computer (i.e. dynamic memory element), and power systems.
- A capacitor consists of two conducting plates separated by an insulator or dielectric.
- Generally the plates are made of aluminum foil
- The dielectrics are made of air, ceramic, paper, or mica.
- When a voltage source (v) is connected to the capacitor, the source deposits a positive charge ' q ' in one plate and a negative charge ' $-q$ ' on the other.
- The stored charge is directly proportional to the applied voltage ' v '.

$$q = Cv$$

Where, C = proportionality constant known as capacitance of the capacitor. The unit of capacitance is farad (F)



Capacitor contd...

- Capacitance is the ratio of the charge on one plate of capacitor to the voltage difference between the two plates, measured in farads (F). 1 farad= 1 coulomb/volt. General range of capacitor in the form of picofarad(pF) or microfarad.

On other word, capacitance is the amount of charge stored per plate for a unit voltage difference in a capacitor.

- Although the capacitance C of a capacitor is the ratio of the charge q per plate to the applied voltage v , it does not depends on it.
- The value of capacitance depends on the physical dimensions of the capacitor.

$$C = \frac{\epsilon A}{d}$$

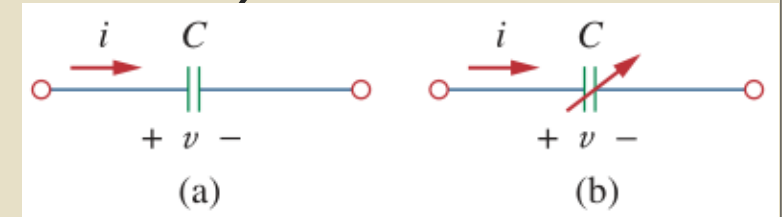
Where A is the surface area of each plate, d is the distance between the plates, ϵ is the permittivity of dielectric material.

Three factors determine the value of the capacitance:

1. The surface area of the plates.
2. The spacing between the plates
3. The permittivity of the material.

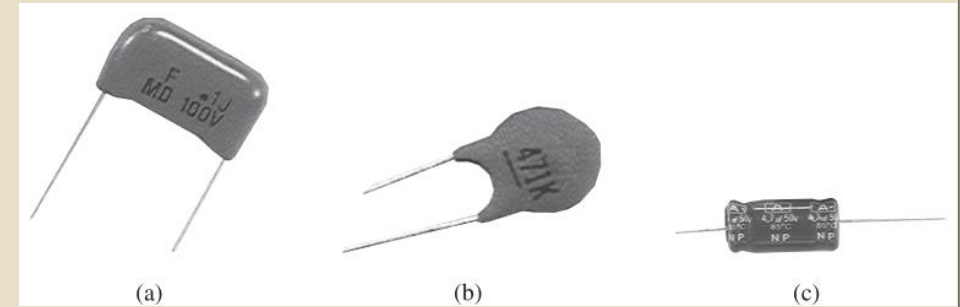
Capacitor contd... (Not for exam)

- Generally fixed capacitor is used in practical purposes.
- The capacitance of a trimmer (or padder) capacitor is often placed in parallel with another capacitor so that the equivalent capacitance can be varied slightly.
- The capacitance of the variable air capacitor (meshed plates) is varied by turning the shaft.
- Variable capacitors are used in radio receivers allowing one to tune to various stations.
- In addition, capacitors are used to block dc, pass ac, shift phase, store energy, start motors, and suppress noise.



Fixed Capacitor

Variable Capacitor



a) Polyester capacitor, b) ceramic capacitor, c) electrolytic capacitor.

Capacitor contd...

We know, current is the rate of change of charge through a specific cross-sectional area (flow of charge in a unit area).

$$\text{Thus, } i = \frac{dq}{dt} = \frac{d(Cv)}{dt} = C \frac{dv}{dt}$$

Capacitors that satisfy the above equations are linear capacitors.

From the above equation we get,

$$v(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + v(t_0)$$

We know, Instantaneous power $p = vi = Cv \frac{dv}{dt}$

Thus, Energy stored in the capacitor, $w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$

Conclusion: This equation represents the energy stored in form of electric field that exists between the plates of the capacitor. This energy can be retrieved, since an ideal capacitor cannot dissipate energy.

The word capacitor is derived from this element's capacity to store energy in an electric field.

Capacitor contd...

The term that resist the current to flow in capacitor is known as capacitive reactance (X_C).

For a capacitor, $X_C = \frac{1}{2\pi fC}$ (At normal frequency)

Where, f = frequency of the input source; C = capacitance of the capacitor.

In case of DC, frequency of the source is zero. Thus $X_C = \frac{1}{2\pi \times 0 \times C} = \infty$

Thus, the circuit acts as open circuit.

In case of Very high frequency, (say $f = \infty$), capacitive reactance becomes $X_C = \frac{1}{2\pi \times \infty \times C} = 0$

Note: The voltage on a capacitor cannot change abruptly.

Problems (Text book- Page 220-222)

1. a) Calculate the charge stored on a 3-pF capacitor with 20 V across it.
b) Find the energy stored in the capacitor.
2. What is the voltage across a 4.5 μF capacitor if the charge on one plate is 0.12 mC? How much energy is stored?
3. The voltage across a 5 μF capacitor is $v(t) = 10 \cos(6000t)$ V, calculate the current through it.
4. If a 10 μF capacitor is connected to a voltage source with is $v(t) = 10 \cos(2000t)$, determine current through the capacitor.
5. Determine the voltage across a 2 μF capacitor if the current through it is $i(t) = 6e^{-3000t}$. Assume that the initial capacitor voltage is zero.
6. The current through a 100- μF capacitor is $i(t) = 50 \sin 120\pi t$ mA. Calculate the voltage across it at $t = 1$ ms and $t = 5$ ms. Take $v(0) = 0$.
Answer: 93.14 mV, 1.736 V.
7. Example 6.4, Practice problem 6.4, Example 6.5, Practice Problem 6.5.

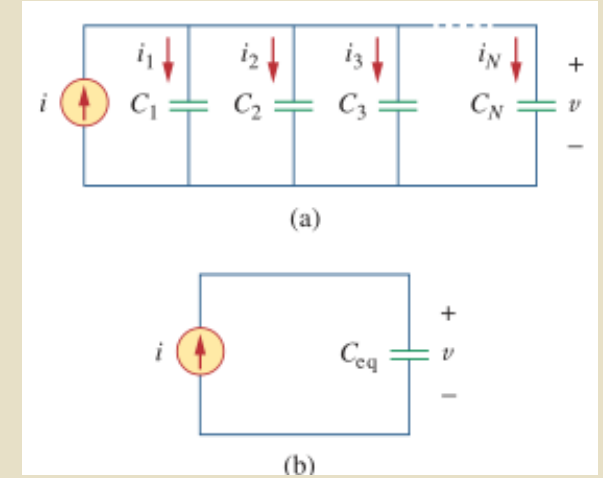
Capacitor in Series/ Parallel

Parallel:

If $C_1, C_2, C_3, \dots, C_N$ are capacitors connected in parallel combination and C_{eqp} is their equivalent capacitance then, $C_{eqp} = C_1 + C_2 + C_3 + \dots + C_N$

Thus, the equivalent capacitance of N parallel connected capacitors is the sum of the individual capacitances.

To know how, see page 222 of textbook.



- a) Parallel-connected N -Capacitors
- b) Equivalent circuit of (a)

Series:

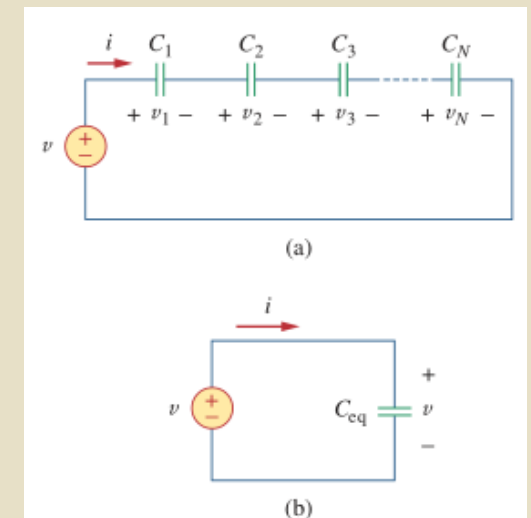
If $C_1, C_2, C_3, \dots, C_N$ are capacitors connected in series combination and C_{eqs} is their equivalent capacitance then, $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$

Thus, the equivalent capacitance in series connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.

To know how, see page 223 of textbook.

We can rewrite the equation as (for two capacitor),

$$C_{eqs} = \frac{C_1 C_2}{C_1 + C_2}$$



- a) Series-connected N -Capacitors
- b) Equivalent circuit of (a)

Problems

1. Find the equivalent capacitance seen between terminals a and b of the circuit shown in fig. 1.
2. Find the equivalent capacitance seen at the terminals of the circuit in fig. 2.
3. Find the voltage across each capacitor of fig. 3.
4. Find the voltage across each of the capacitors in fig. 4.

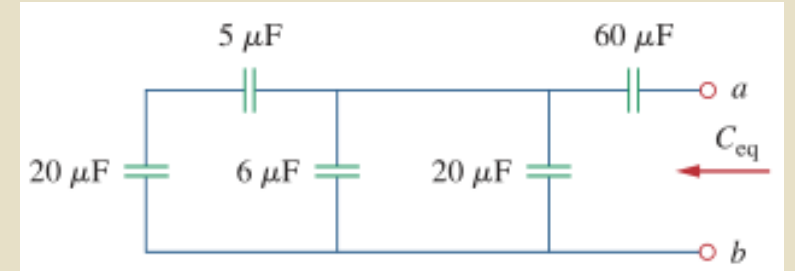


fig. 1

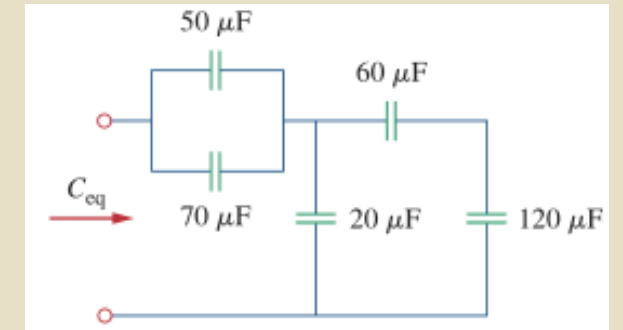


fig. 2

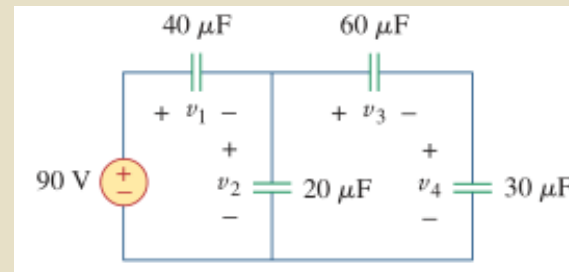


fig. 4

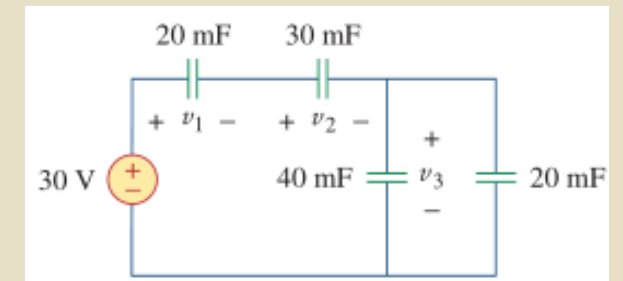


fig. 3

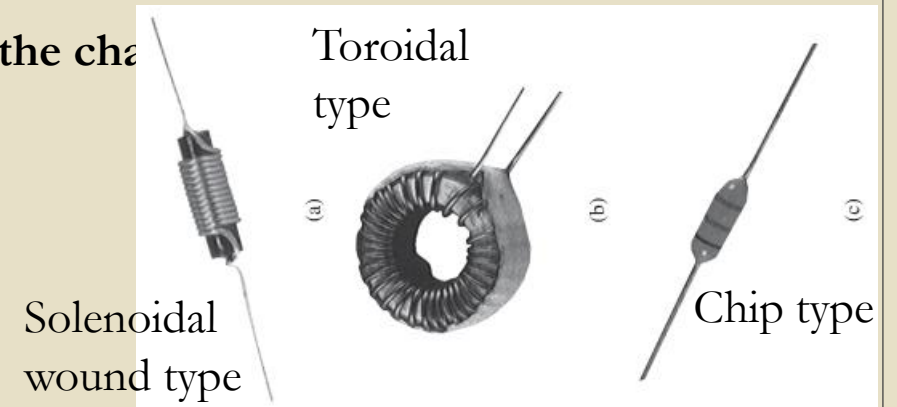
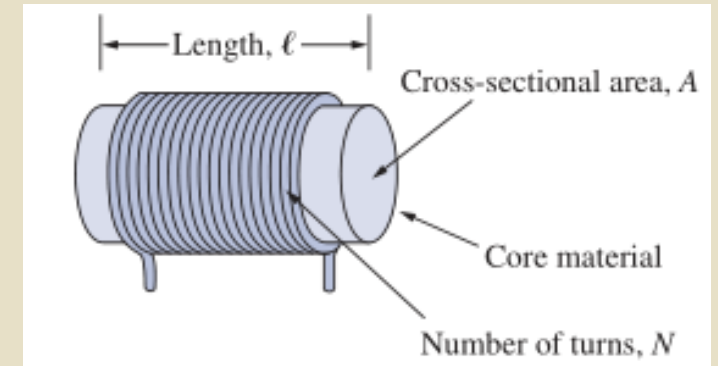
Inductors

- A passive element designed to store energy in its magnetic field.
- Uses: Electronics (radios, TVs, radars) and power system (Power supplies, transformers, electric motors).
- Any conductor of electric current has inductive properties and may be regarded as inductor. To enhance its inductive effect, a practical inductor is usually formed into a cylindrical coil with many turns of conducting wire.
- “An Inductor consists of a coil of conducting wire”
- If current is pass through an inductor, the expression of voltage becomes

$$v = L \frac{di}{dt}$$

- Where, L= Inductance. The unit of inductance is henry (H). $1H = 1 \frac{\text{Volt-second}}{\text{ampere}}$

“ Inductance is the property whereby an inductor exhibits opposition to the change in current flowing through it, measured I henrys.”



Inductance Contd...

- The inductance of an inductor depends on its physical dimension and construction, that is derived from electromagnetic theory. For example, For solenoidal inductor, $L = \frac{N^2 \mu A}{l}$ (Where, N=Number of turns, l=length, A= Cross-sectional area, μ = permeability of the core).
- Range of inductor values (few microhenry's in case of communication system, tens of henrys in case of power systems)
- Inductor value may be fixed or variable.
- The core may be made of iron, steel, plastic, or air.
- The terms coil and choke are also used for inductors.
- If $v(t)$ voltage is applied across an inductor, current through it $i(t)$ is,

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

Energy stored, $w(t)$ is

$$w = \frac{1}{2} Li^2$$

N.B: To know more about the derivation see page 227 of the text book.

Voltage current relationship shows that voltage changes linearly with the rate of change of current

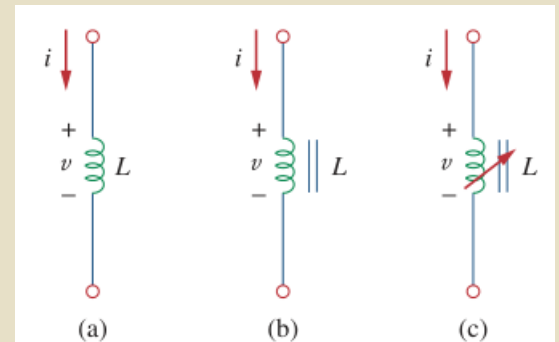


Figure 6.23

Circuit symbols for inductors: (a) air-core, (b) iron-core, (c) variable iron-core.

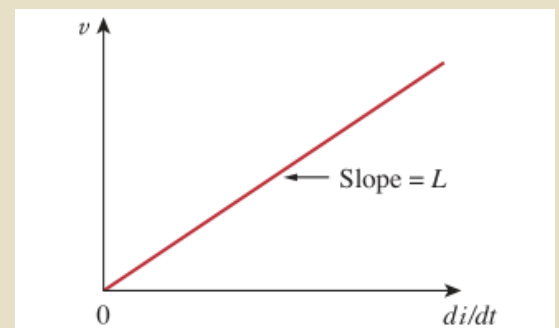


Figure 6.24

Voltage-current relationship of an inductor.

Inductive reactance (X_L)

In normal case, If the source frequency is f , then the inductive capacitance $X_L = 2\pi fL$

In case of DC source:

$f=0$, thus $X_L = 2\pi \times 0 \times L = 0 \Omega$

So, inductors act as short circuit in case of DC.

In case of very high frequency (say, infinity):

$f = \infty$, thus $X_L = 2\pi \times \infty \times L = \infty \Omega$

So, inductors act as open circuit in case of very high frequency.

Note that, the current through an inductor cannot change instantaneously. Ideal inductor does not dissipate energy, it stores energy that can be retrieved later.

Problems

Example 6.8

The current through a 0.1-H inductor is $i(t) = 10te^{-5t}$ A. Find the voltage across the inductor and the energy stored in it.

If the current through a 1-mH inductor is $i(t) = 60 \cos 100t$ mA, find the terminal voltage and the energy stored.

Practice Problem 6.8

Answer: $-6 \sin 100t$ mV, $1.8 \cos^2(100t)$ μ J.

Find the current through a 5-H inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t < 0 \end{cases}$$

Also, find the energy stored at $t = 5$ s. Assume $i(v) > 0$.

The terminal voltage of a 2-H inductor is $v = 10(1 - t)$ V. Find the current flowing through it at $t = 4$ s and the energy stored in it at $t = 4$ s. Assume $i(0) = 2$ A.

Example 6.9

Example 6.10

Consider the circuit in Fig. 6.27(a). Under dc conditions, find: (a) i , v_C , and i_L , (b) the energy stored in the capacitor and inductor.

Practice Problem 6.10

Determine v_C , i_L , and the energy stored in the capacitor and inductor in the circuit of Fig. 6.28 under dc conditions.

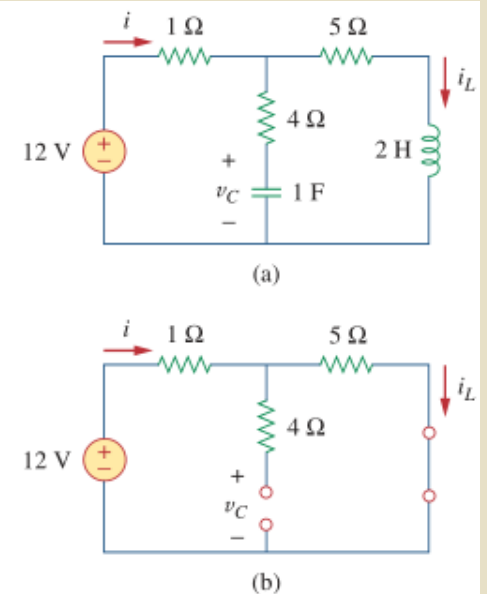


Figure 6.27
For Example 6.10.

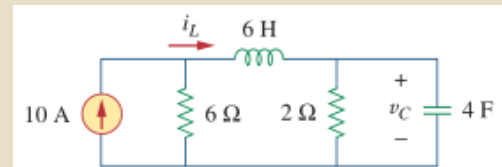


Figure 6.28
For Practice Prob. 6.10.

Inductors (in Series and Parallel)

Series connected inductors:

If $L_1, L_2, L_3, \dots, L_N$ are inductors connected in series. Then equivalent inductance L_{eqs} will be,

$$L_{eqs} = L_1 + L_2 + L_3 + \dots + L_N$$

“The equivalent inductance of series-connected inductors is the sum of the individual Inductances”

To know the derivation, see page 230 of the text book.

Parallel connected inductors:

If $L_1, L_2, L_3, \dots, L_N$ are inductors connected in parallel. Then equivalent inductance L_{eqp} will be,

$$\frac{1}{L_{eqp}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N}$$

“The equivalent inductance of parallel inductors is the reciprocal of the sum of the reciprocals Of the individual inductances.”

For two inductors in parallel,

$$L_{eqp} = \frac{L_1 \times L_2}{L_1 + L_2}$$

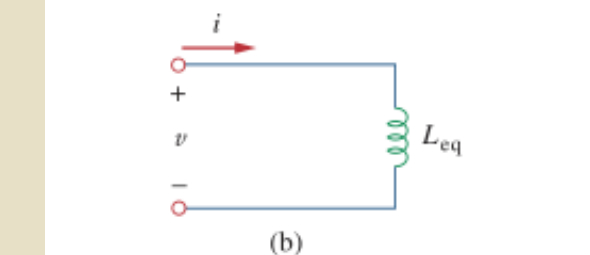
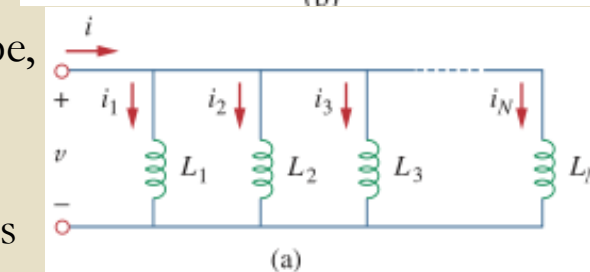
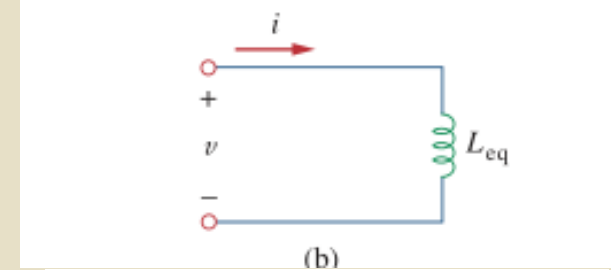
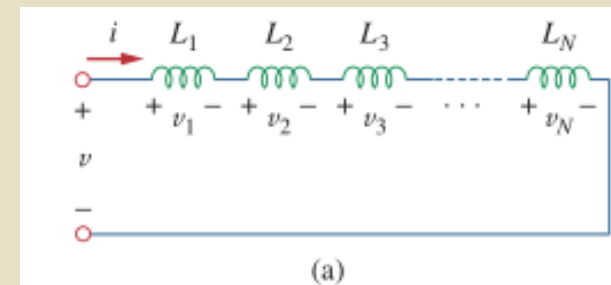


TABLE 6.1

Important characteristics of the basic elements.[†]

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v - i :	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$	$v = L \frac{di}{dt}$
i - v :	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{\text{eq}} = R_1 + R_2$	$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{\text{eq}} = L_1 + L_2$
Parallel:	$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{\text{eq}} = C_1 + C_2$	$L_{\text{eq}} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

Problems

Example 6.11

Find the equivalent inductance of the circuit shown in Fig. 6.31.

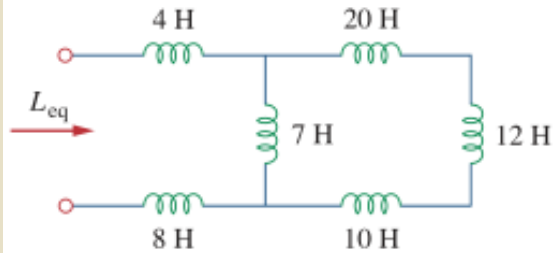


Figure 6.31

For Example 6.11.

Solution:

The 10-H, 12-H, and 20-H inductors are in series; thus, combining them gives a 42-H inductance. This 42-H inductor is in parallel with the 7-H inductor so that they are combined, to give

$$\frac{7 \times 42}{7 + 42} = 6 \text{ H}$$

This 6-H inductor is in series with the 4-H and 8-H inductors. Hence,

$$L_{eq} = 4 + 6 + 8 = 18 \text{ H}$$

Practice Problem 6.11

Calculate the equivalent inductance for the inductive ladder network in Fig. 6.32.

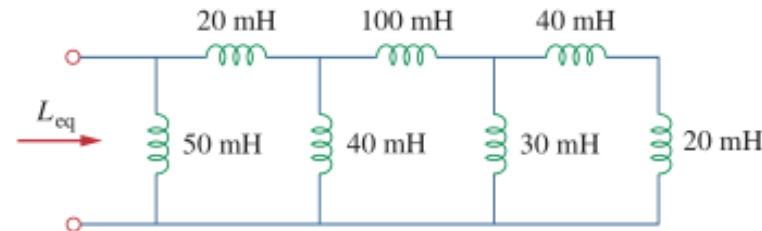


Figure 6.32

For Practice Prob. 6.11.

Answer: 25 mH.

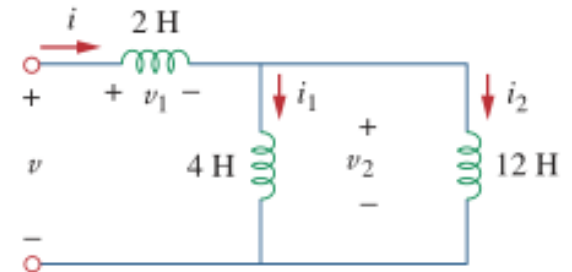


Figure 6.33

For Example 6.12.

For the circuit in Fig. 6.33, $i(t) = 4(2 - e^{-10t})$ mA. If $i_2(0) = -1$ mA, find: (a) $i_1(0)$; (b) $v(t)$, $v_1(t)$, and $v_2(t)$; (c) $i_1(t)$ and $i_2(t)$.

Example 6.12

But in science the credit goes to the man
who convinces the world, not to the man to
whom the idea first occurs.

Francis Darwin